EVNet Sentinel: A Leakage-Corrected Reproduction and Comparative Study of Intrusion Detection on Electric Vehicle Charging Networks

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***Abstract*—Public electric vehicle chargers are networked computers, and an attack on their link to the central management system can stop charging and billing. Machine-learning intrusion detectors evaluated on the CICEVSE2024 dataset report accuracies near 99 percent. This study reproduces the most recent of them, an online Adaptive Random Forest with ADWIN drift detection, and audits whether that accuracy reflects attack detection. We replay the published preprocessing, probe every feature for leakage, and retrain six static classifiers and the online model on corrected data over three seeds, reporting macro-F1. A single capture timestamp recovers the attack label with 99.76 percent accuracy, above the published 98.40 percent, and the drift detector fires on the joins between capture files rather than on changes in traffic. After correction, flooding attacks remain near-perfectly separable while six network-scan types are not, and a mutual-information analysis shows that this limit lies in the flow features rather than in the models. A per-attack test finds that no attack passes as normal traffic; the tree ensembles are statistically tied, and XGBoost is selected as the primary detector by its marginally lower false-alarm rate. Because detection alone does not stop an attacker, we close the loop with an operator-approved containment policy matched to how reliably each attack is named: quarantine for floods, restriction for scans, and a stopped session for a compromised vehicle, with escalation when a device that scanned begins to flood. Replayed over every recorded attack, the policy contains all 20 attacker–attack pairs and takes no action on benign traffic. The work is delivered as EVNet Sentinel, with a prediction service, the response policy and an interactive web front end.**

***Index Terms*—Intrusion detection, electric vehicle charging, CICEVSE2024, data leakage, concept drift, OCPP, incident response, machine learning evaluation**

I. Introduction

Electric mobility depends on a growing network of public charging stations (Electric Vehicle Supply Equipment, EVSE). Each station talks to the vehicle over ISO 15118 and to a Charging Station Management System (CSMS) over the Open Charge Point Protocol (OCPP) [14]. The OCPP link carries authorisation, metering and remote commands, so an attacker who floods or probes a station can interrupt charging and the revenue that depends on it [1], [3]. Detecting such attacks on the station network is therefore a practical security need.

Most detectors for this setting are machine-learning classifiers trained on the CICEVSE2024 dataset [8], and reported accuracies are close to perfect [1], [2]. Makhmudov et al. [4] proposed the first online detector for charging networks, combining an Adaptive Random Forest (ARF) [6] with ADaptive WINdowing (ADWIN) [7] so the model can adapt as traffic changes, a problem known as concept drift [15]. They report 98.40% multiclass accuracy.

Near-perfect scores on a public benchmark deserve scrutiny. Security datasets are often recorded one attack per capture, and features that identify the recording rather than the traffic can leak the label [10], [11], [13]. **The research problem** addressed here is whether the published accuracy measures attack detection or a property of how CICEVSE2024 was assembled. The problem is significant because an operator who trusts an inflated score will deploy a detector that fails on the attacks that matter.

Objectives. The study aims to (i) reproduce the published pipeline and test it for leakage; (ii) report corrected, multi-seed benchmarks of static and online models; (iii) explain which attacks remain hard and why; (iv) choose a primary detector from detection, naming and false-alarm rates; (v) turn detections into contained attacks; and (vi) deliver a usable system around the corrected models.

Research questions. RQ1: does the published preprocessing leak the label? RQ2: is the reported drift real? RQ3: what

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accuracy remains, and which attacks are hard? RQ4: which model should an operator trust? RQ5: can an operator act on the detections, including scans that cannot be named?

Contributions. We present EVNet Sentinel, a passive network intrusion detection system for the station network, and use it to deliver five verifiable leakage findings, corrected benchmarks, a model-independent explanation of the reconnaissance ceiling, a per-attack evaluation with a confidence-gated naming rule, a closed-loop response policy evaluated by replay over every recorded attack, and a working prediction service and web front end.

II. Literature Review

Surveys and threat analyses. Bean et al. [1] review 24 CICEVSE2024 studies and name concept drift as an open problem, but take reported accuracies as given; Malimage et al. [3] analyse incidents and protocol weaknesses without proposing a detector.

Detectors on CICEVSE2024. Joglekar et al. [2] combine a network detector (XGBoost, 99.99% accuracy) with a host detector. They remove IP and MAC identifiers but keep timestamps, and train offline. Makhmudov et al. [4] stream the data through ARF+ADWIN with 20 trees and $\delta = 0.002$, reporting 99.13% binary and 98.40% multiclass accuracy, 11 drift events and 3.7 ms per flow; drift handling is offered as the main advantage. Tanyildiz et al. [5] instead predict the time to the next attack with a GAN-augmented regressor on a different dataset.

Online learning and drift. Gama et al. [15] stress that a detector should be judged against the change it is meant to find. ARF [6] gives each tree its own drift detector, and ADWIN [7] cuts a window of recent errors when two sub-windows differ; River [9] provides both, and Random Forest [16] is the offline reference.

Evaluation pitfalls. Kaufman et al. [10] formalise leakage as information in the training data that would not be available at prediction time. Arp et al. [11] list sampling bias, spurious correlations and inappropriate baselines among the most common errors in security machine learning, and Sommer and Paxson [13] warn that intrusion detectors often learn artefacts of the evaluation environment. **Research gap.** No prior CICEVSE2024 study tests its preprocessing for capture-level leakage, reports results over several seeds, uses a class-balanced metric as the headline, or checks whether detected drift is real. None evaluates what happens after a detection. This study addresses all five gaps and adds a per-attack, operator-oriented evaluation.

III. Proposed Methodology and Architecture

Fig. 1 summarises the method and the delivered system, from capture to response.

A. Research Design

The study is *quantitative and computational*, in two parts. The first is a *reproduction audit*: the published pipeline is replayed exactly and each suspected artefact is isolated with a controlled experiment that changes one factor at a time (a feature, the stream order, or the split). The second is a *comparative experiment*: six static classifiers and the online model are trained on corrected data under one protocol. This design suits the objectives because each finding becomes a measurable, repeatable test rather than an opinion, and every number can be regenerated from the public code.

The detector is *passive*. It reads a mirrored copy of the switch traffic, classifies each completed network flow and raises an alert, but never blocks traffic. This matches how CICEVSE2024 itself was recorded.

B. Data

CICEVSE2024 [8] was recorded by the Canadian Institute for Cybersecurity on a testbed with two charging stations (EVSE-A and EVSE-B), a local and a remote CSMS, and a vehicle controller, with stations either idle or charging. Network flows were extracted with NFStream into 86 features and labelled from the capture file name. The 59 capture files hold 2,744,700 flows in 15 classes:

- five *volumetric floods*: SYN, TCP, UDP, PSH-ACK and SynonymousIP;
- six *nmap reconnaissance modes*: TCP port scan, SYN stealth scan, OS fingerprinting, service-version detection, vulnerability scan and aggressive scan;
- three *low-rate attacks*: ICMP flood, ICMP fragmentation and Slowloris;
- benign traffic.

The dataset lists its devices but not which device attacked which. Because every flow carries source and destination MAC addresses, we derived the route of each attack capture: a Kali PC on the Wi-Fi attacks both stations, a Raspberry Pi sends the UDP floods, and a compromised vehicle scans EVSE-B. Every attack targets a charging station, and in six captures the compromised vehicle scans EVSE-B over the charging cable. The classes are highly imbalanced: after deduplication the four large floods hold about 196,000–260,000 flows each, the scans 28,000–39,000, Slowloris 2,721, and benign traffic only 81.

C. Materials and Tools

All experiments ran on one Apple M5 laptop (10 cores, 16 GB RAM) with Python 3.13, scikit-learn 1.8 [12], XGBoost, LightGBM and River 0.26 [9]; a FastAPI service and a Next.js 16 front end deliver the models. Code and results are public (Section VI).

D. Procedure

Step 1, replay. The reference preprocessing [4] was rerun verbatim to reproduce its feature set.

Step 2, leakage probes. A depth-25 decision tree was trained on each feature alone and scored by 3-fold cross-validated accuracy. Keeping or dropping each suspicious column was crossed before deduplication.

Step 3, corrected preprocessing. All captures are concatenated and labelled. Identifiers (flow ID, IP, MAC, vendor) and constant columns are removed. The six absolute timestamps

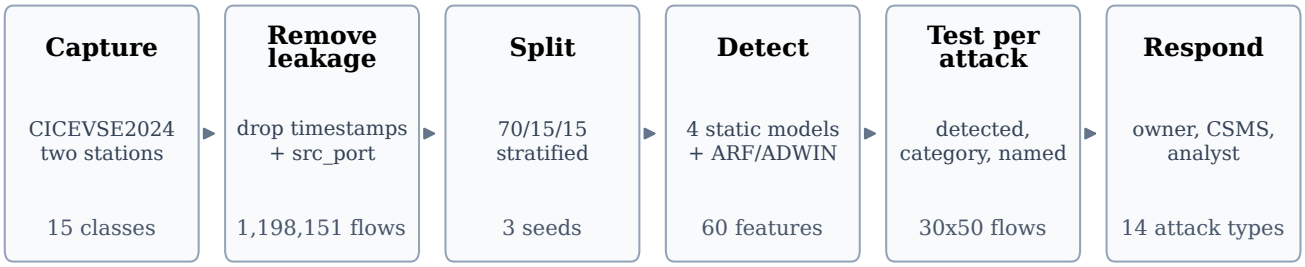


Fig. 1. Method and system at a glance, from capture to response.

(`*_first_seen_ms`, `*_last_seen_ms`) and `src_port` are dropped, while the destination port is kept. Duplicate rows are removed, the data are split 70/15/15 with class stratification, and each feature is standardised with training statistics only. This leaves 1,198,151 flows and 60 features (838,705 / 179,723 / 179,723 rows).

Step 4, training. The static models are:

- Random Forest [16]: 100 trees, depth 15;
- Decision Tree: depth 20;
- XGBoost: up to 400 rounds, depth 8, learning rate 0.1;
- LightGBM: up to 600 rounds, 63 leaves, learning rate 0.05;
- Logistic Regression and a linear SVM, both trained by stochastic gradient descent.

All six use balanced class weights $w_k = N/(Kn_k)$ (as sample weights for the boosted models), so that rare classes count as much as common ones. The boosted models stop early on the validation split, after 110–125 rounds; their settings were fixed in advance, not tuned on the test set. ARF uses the published configuration: 20 trees, half the features per split, a grace period of 30, and ADWIN with $\delta = 0.002$. It is run prequentially, predicting each flow before learning from it, in both capture order and shuffled order.

Step 5, per-attack testing. For each attack, model and seed, 30 batches of 50 unseen flows are scored. Mixed campaigns of four simultaneous attacks are also run. Regression gates fail the run if detection of any attack falls below 0.95.

Step 6, response replay. Every attack class is streamed, from each device that really sent it, through the containment policy of Section III-F; Section IV-F gives the protocol.

Step 7, delivery. The models and the policy are served by the API, and the exported results drive the web dashboard and a simulation that replays held-out flows along their recorded routes.

E. Data Analysis

Metrics. For class k with precision P_k and recall R_k,

$$F_{1,k} = \frac{2P_kR_k}{P_k + R_k}, \qquad \text{macro-}F_1 = \frac{1}{K}\sum_{k=1}^{K} F_{1,k}. \tag{1}$$

Macro-F_1 weights the 15 classes equally; accuracy weights each *flow* equally and is therefore dominated by the floods. After deduplication the floods are 83.5% of our flows, and 99.2% when the published feature set is used with its timestamps removed. On that set Random Forest reaches 0.996 accuracy but only 0.70 macro-F_1: a model that names every flood and misses most scans still scores near-perfect accuracy, which is why macro-F_1 is our headline. Every figure is the mean and standard deviation over seeds 42, 1337 and 2024, where each seed changes both the split and the model initialisation.

Reporting precision. A score below 1 is never rounded up to 1: values in $[0.995, 1)$ are given to four decimals and rounded down if rounding would reach 1. A standard deviation is rounded *up* at the same precision, so a seed-to-seed spread of 3×10^{-5} is reported as ± 0.0001 rather than ± 0.0000.

Stream order. Prequential accuracy is compared with the trivial rule "repeat the previous label". That rule's accuracy equals the label autocorrelation

$$\rho = \Pr\left[y_t = y_{t-1}\right], \tag{2}$$

which any model on an ordered stream must beat before it can be said to learn. For a shuffled stream $\rho = \sum_k p_k^2$, where p_k is the share of class k.

ADWIN. ADWIN signals drift when, for some split of its window W into W_0W_1, the mean error of the two parts differs by more than

$$\epsilon_{\text{cut}} = \sqrt{\frac{1}{2m}\ln\frac{4n}{\delta}}, \qquad m = \frac{1}{1/n_0 + 1/n_1}, \tag{3}$$

where n_0 and n_1 are the sizes of the two parts and $n = |W|$. In capture order every file boundary is an abrupt change by construction, so we test whether the alarms fall on them.

Drift alignment. For each ADWIN alarm e_i, we measure the distance to the nearest capture-file boundary, $d_i = \min_j |e_i - b_j|$. We count the alarms within $w = 100$ flows of a boundary and compare that count H with the counts $H^{(r)}$ from $R = 2000$ random placements of the same number of alarms:

$$p = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\left[H^{(r)} \geq H\right]}{1 + R}. \tag{4}$$

Per-feature separability. Within one category g (DoS or reconnaissance), we measure how much of the class uncertainty a single feature X_j removes:

$$S_j^{(g)} = \frac{I\left(X_j; Y \mid g\right)}{H\left(Y \mid g\right)}, \tag{5}$$

estimated with a k-nearest-neighbour mutual-information estimator [17] on 6,000 held-out flows per category. S_j does not depend on any model, so it describes the data itself.

Per-attack rates. For each batch we measure three rates: the share detected (called any attack), the share placed in the right category, and the share named exactly. On benign flows we measure the false-alarm rate. The primary detector is the model with the fewest false alarms among those whose mean exact-naming rate is within 0.02 of the best. Once a rate of k passes in n tested flows reaches 0.99, it is reported as its Wilson 95% lower bound

$$L = \frac{\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad \hat{p} = \frac{k}{n}, \; z = 1.96, \tag{6}$$

so that 4,500 passes in 4,500 flows reads ≥ 0.9991 rather than a perfect 1, and a perfect run always ranks above one with a miss.

Confidence gating. A model names an attack only if its top class probability is at least t:

$$\text{output}(\mathbf{x}) = \begin{cases} \arg\max_k \hat{p}(k \mid \mathbf{x}) & \text{if } \max_k \hat{p}(k \mid \mathbf{x}) \geq t, \\ \text{category of that class} & \text{otherwise.} \end{cases} \tag{7}$$

F. Closed-loop Response

Detection alone does not stop an attacker, so the system closes the loop: *detect, contain, harden, verify*. A site threat level is kept over the last 20 judged flows: the site is *elevated* once 25% are flagged and *unsafe* at 60%, and it steps down only after 10 quiet flows. On top of it, the policy recommends one containment action per attacking device, matched to how reliably that attack is named (Section IV-E):

- **Quarantine** (switch port or ACL): the site is unsafe and one device has sent at least 5 denial-of-service flows at confidence ≥ 0.9. Floods are named almost perfectly, so this evidence suffices to cut the device off.
- **Restrict** (no access to the stations or the CSMS; Internet kept): the site is at least elevated and one device has sent at least 5 flows *recognised as reconnaissance*. Scans are placed in the right category reliably but rarely named, so the rule uses the category only and stops short of a quarantine. The device is then *watched*.
- **Stop session**: the scanning device is the vehicle itself. Its traffic never crosses the site switch, so the station ends the session (OCPP RemoteStopTransaction) and blocks the vehicle's identifier.
- **Escalation**: a watched device that sends at least 2 denial-of-service flows at confidence ≥ 0.6 is recommended for quarantine at once.

TABLE I
LABEL RECOVERED FROM A SINGLE FEATURE (15 CLASSES, 3-FOLD CV)

Feature	Nature	Accuracy
`bidirectional_first_seen_ms`	fingerprint	**0.9976**
`src_port`	partial	0.4028
`dst_port`	legitimate	0.2917
Published multiclass accuracy [4]		*0.9840*

A person approves every action, and stations and the CSMS can never be contained. Each action comes with role-specific hardening steps (station owner, CSMS operator, network analyst) so that the same attack fails next time, and verification counts the attack flows stopped after containment.

IV. RESULTS

A. Leakage in the published pipeline

Table I lists the leakage probes. A decision tree on `bidirectional_first_seen_ms` alone recovers the 15-class label with 0.9976 accuracy, above the 0.9840 reported for the full published model. The reference preprocessing keeps all six timestamp columns. It also drops `dst_port`, the feature that defines a port scan. The source port, which the operating system allocates almost sequentially, alone recovers 0.4028.

The published configuration is also fragile. Dropping the destination port while keeping the timestamps leaves 273,932 reconnaissance flows after deduplication. These flows can then be separated by *when* they were captured rather than by what they did. Removing both the timestamps and the destination port collapses the scans to 6,821 flows: almost every remaining scan flow becomes a duplicate.

B. Stream order and drift

In capture order each file holds a single class, so over 1,921,182 training flows the label changes only 54 times and $\rho = 0.99997$. ARF+ADWIN reaches 0.9997, which does not beat the repeat-last rule. On a shuffled stream it reaches 0.5585. Of the 43 drift alarms in capture order, 42 lie within 100 flows of a file boundary, with a median distance of 22 flows. Random placement would put only 2.2 ± 1.5 alarms that close ($p < 0.0005$).

C. Corrected benchmarks

Table II gives the corrected results. The four tree-based models, Random Forest, Decision Tree, XGBoost and LightGBM, are within one standard deviation of each other: gradient boosting does not break the ceiling. The linear models trail by more than 0.1 macro-F_1, and all six reach about 0.86 accuracy. On a like-for-like split, removing the timestamps lowers Random Forest's macro-F_1 from 0.9547 to 0.5894, while the SVM changes by only $+0.03$. Removing `src_port`, with the rows and the split held fixed, costs the deeper Decision Tree 0.170 macro-F_1, twice as much as Random Forest (0.076). Extra depth was buying finer bands of port numbers, so the tree's earlier advantage was memorisation.

TABLE II
MULTICLASS RESULTS, MEAN ± STD OVER 3 SEEDS (CORRECTED FEATURES)

Model	Macro-F_1	Acc.	Fit (s)
Random Forest	**0.558±0.008**	0.868	8.5
Decision Tree	0.557±0.010	0.868	3.4
XGBoost	0.551±0.011	0.869	33.0
LightGBM	0.550±0.004	0.868	42.4
Logistic Regression	0.436±0.017	0.863	78.9
Linear SVM	0.411±0.017	0.862	34.9

TABLE III
PER-CLASS F_1, RANDOM FOREST, MEAN ± STD OVER 3 SEEDS

Class	Support	F_1
SYN, TCP, PSH-ACK, SynIP	196k–259k	0.9999±0.0001
UDP flood	32,072	0.9994±0.0001
TCP port scan	35,123	0.265
OS fingerprinting	27,702	0.223
Vulnerability scan	39,002	0.204
Service-version detection	30,338	0.156
Aggressive scan	27,815	0.145
SYN stealth scan	34,765	0.105
Benign	81	0.986
ICMP flood / ICMP fragmentation	32 / 28	0.660 / 0.439
Slowloris	2,721	0.191

D. Per-class results and separability

Table III shows a sharp split. The five floods reach $F_1 \geq 0.999$ in every seed. The six scans average 0.183, and sample size does not explain the gap: UDP flood (32,072 flows) scores 0.9994, while TCP port scan (35,123 flows) scores 0.265. The separability measure (5) shows the same split. Within DoS, one inter-arrival-time feature resolves 98% of the class uncertainty. Within reconnaissance, the best feature resolves 5%.

The errors among the scans follow a pattern. Random Forest labels 32–48% of every other scan as OS fingerprinting, so the scans collapse into one "sink" class. A further 8–11% of each scan is labelled Slowloris, because both hold long-lived, nearly idle connections. This confusion accounts for almost all of the error between the reconnaissance and DoS categories.

E. Per-attack detection and confidence

Table IV lists the exact-naming rate per attack. Across 90 mixed campaigns, every model flags at least 99.9% of attack flows as malicious. Every model names the floods exactly (Wilson lower bound ≥ 0.977 for every model, ≥ 0.998 for Random Forest). The scans are placed in the right category 96% of the time by the best model for each, but named exactly only 35% of the time. Table V gives the false-alarm rates on benign traffic. By the selection rule of Section III, XGBoost is the primary detector; its lead over Random Forest (0.02 against 0.03) is within the 95% intervals of the 12 benign test flows, so the two are interchangeable in practice. Applying the confidence gate (7) at $t = 0.5$ cuts Random Forest's wrong scan names from 81% to 3% and reports 94% of scans by category, while 93% of DoS flows are still named exactly.

TABLE IV
EXACT-NAMING RATE E PER ATTACK FOR RANDOM FOREST, THE BEST MODEL AT NAMING, AND THE BEST CATEGORY RATE C (RATES ≥ 0.99 AS WILSON 95% LOWER BOUNDS, $n = 4{,}500$)

Attack	RF E	Best E	Best C
Five volumetric floods	≥0.998	all (≥0.977)	≥0.999
Slowloris	0.82	XGB 0.83	0.83
TCP port scan	0.33	LR 0.66	0.97
OS fingerprinting	0.42	XGB 0.43	0.95
SYN stealth scan	0.08	SVM 0.43	0.97
Vulnerability scan	0.14	SVM 0.26	0.97
Aggressive scan	0.12	LGBM 0.17	0.94
Service-version det.	0.10	DT 0.13	0.97

TABLE V
FALSE-ALARM RATE ON BENIGN TRAFFIC BY MODEL

	RF	DT	XGB	LGBM	LR	SVM
False alarms	0.03	0.11	**0.02**	0.03	0.44	0.58

TABLE VI
CONTAINMENT REPLAY (MEANS OVER 20 INTERLEAVINGS PER PAIR): ATTACK FLOWS SEEN BEFORE AN ACTION IS RECOMMENDED, AND ATTACK FLOWS THAT REACHED THE STATION BEFORE IT TOOK EFFECT

Attack (pairs)	Action	Contained	Before	Through
DoS (8)	Quarantine	8/8	12.0	14.6
Scans, Wi-Fi (6)	Restrict	6/6	5.5	8.8
Scans, vehicle (6)	Stop session	6/6	7.7	10.7

Escalation: a scanner that floods is quarantined after 1 flow (11 cold)
Benign only (2,000 flows): 0 actions

F. Closed-loop response, replayed

The policy of Section III-F was replayed over every attack class, from each device that really sent it (20 attacker–attack pairs). Each run streams 20 benign flows, then 120 flows of which 70% are that device's held-out attack flows, through XGBoost's recorded verdicts; a simulated operator approves each recommendation 5 flows after it appears, after which the device's attack flows are dropped before reaching the station. Each pair is run with 20 random interleavings. Table VI summarises the result.

Every pair is contained, each with the action intended for it: no scanning device is ever quarantined on a scan alone, and the vehicle's scans always lead to a stopped session rather than a switch block. A device that has scanned is quarantined on its first flood flow, against 11 flows when it floods without warning. On benign traffic alone the policy never acts. Three caveats apply. This is a replay of recorded traffic through the policy, not live blocking. The dataset holds only 12 distinct held-out benign flows, so the zero is a small sample. And ICMP flood, ICMP fragmentation and the vehicle's scans have 4–26 held-out flows each, which the replay reuses.

V. DISCUSSION

What the published accuracy measures. Each attack was recorded in its own file at its own time, so an absolute

timestamp identifies the recording, and through it the label. This is leakage in the sense of Kaufman et al. [10]: the feature would not carry this information on live traffic. Trees exploit it and linear models cannot, which is why removing the timestamps costs Random Forest 0.365 macro-F_1 but leaves the SVM almost unchanged. Much of the reported gap between tree and linear models in earlier work therefore reflects how well each model could use the leak.

Drift. Capture-order accuracy equals the repeat-last baseline and the alarms sit on file boundaries: the stream changes where files were joined, not because traffic evolved, so CICEVSE2024 cannot test the drift handling claimed in [4] (cf. [11], [13]).

Floods and scans. The near-perfect flood scores are plausible rather than leaked. The fingerprint columns are removed; a flood has a strong physical signature that a single timing feature captures; and the linear models also name the floods. We nevertheless claim this only for this testbed: the random split can place near-identical flows from one capture on both sides of the split, and floods sent at a lower rate would look different. The scans are a different matter. All six are modes of one tool, the aggressive mode includes two of the others, and no single flow statistic separates them (best $S_j = 0.05$). More model capacity cannot recover information the features do not contain.

Operational meaning. No attack passes as normal traffic: the weakness is *naming*, not *noticing*. XGBoost is selected by the pre-stated rule, though Random Forest trains four times faster and is equally sound. For scans, a confidence gate turns confidently wrong names into correct category alerts, which is the information an operator can act on. The response policy uses exactly that information: it restricts a scanning device on the category alone, keeps quarantine for the attacks that are named reliably, and lets an earlier scan lower the bar for the attack that follows. Detection quality thus decides the *strength* of the response, not whether there is one.

Limitations. Labels come from file names; only 81 benign flows exist, so false-alarm rates are indicative; the split is random rather than grouped; the study covers one testbed; and the response is evaluated by replay, not on live equipment.

VI. Conclusion

This study set out to test whether the high accuracies reported on CICEVSE2024 measure attack detection, and found that they largely do not:

- one capture timestamp beats the published model;
- streaming accuracy in capture order is matched by repeating the previous label;
- the drift detector responds to file boundaries.

With these effects removed, the corrected benchmark is 0.558 macro-F_1 for Random Forest over three seeds. Volumetric floods remain near-perfectly separable on this testbed, while the six nmap scan modes are not, and a model-independent analysis places that limit in the flow features. Gradient-boosted trees (XGBoost, LightGBM) tie with Random Forest rather than surpass it. Operationally, every attack is detected, XGBoost is the recommended detector (statistically tied with Random Forest), and requiring confidence before naming an attack replaces most wrong scan names with correct category alerts. Closing the loop with an operator-approved policy contains every recorded attacker–attack pair in replay, without acting on benign traffic. The system, code and results are public at https://github.com/Mukta01/EVNet_Sentinel.

VII. Future Work

Results should be reported with whole capture files held out, and packet-sequence features tested on the scan modes, which flow statistics cannot separate. The containment policy should be validated on a live OCPP testbed with real switch ACLs and session control, and with far more benign traffic. Dataset builders should release per-capture identifiers and benign traffic in volume.

Acknowledgment

We thank the Canadian Institute for Cybersecurity for CICEVSE2024.

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